MODULE *Exokernel*

EXTENDS *Naturals*, *Sequences*

Parameters and constants

CONSTANT *TaskCount*, *ResourceCount*

VARIABLE *tasks*, *resources*, *allocations*

TaskId are modeled as a sequence of task *IDs*, where each task has an index.

$TaskId \triangleq 1 \,..\, TaskCount$

ResourceId are modeled as a sequence of resource *IDs*, where each resource has an index.

$ResourceId \triangleq 1 \,..\, ResourceCount$

$TaskState \triangleq \{\text{"waiting"}, \text{"running"}\}$

$Availability \triangleq \{\text{"free"}, \text{"busy"}\}$

Type invariant of the specification

$ExokernelTypeInvariant \triangleq$
$\quad \land\ tasks \in [TaskId \rightarrow TaskState]$
$\quad \land\ resources \in [ResourceId \rightarrow Availability]$
$\quad \land\ allocations \in [TaskId \rightarrow [ResourceId \rightarrow \mathrm{BOOLEAN}]]$

Define *Exokernel* state as a record containing tasks, resources, and allocations

$ExokernelInit \triangleq$
$\quad \land\ tasks = [t \in TaskId \mapsto \text{"waiting"}]$
$\quad \land\ resources = [r \in ResourceId \mapsto \text{"free"}]$
$\quad \land\ allocations = [t \in TaskId \mapsto [r \in ResourceId \mapsto \mathrm{FALSE}]]$

A task can request access to a resource if it's not already in use.

$RequestResource(t, r) \triangleq$
$\quad \land\ resources[r] = \text{"free"}$
$\quad \land\ allocations[t][r] = \mathrm{FALSE}$
$\quad \land\ tasks[t] = \text{"waiting"}$
$\quad \land\ tasks' = [tasks\ \mathrm{EXCEPT}\ ![t] = \text{"running"}]$
$\quad \land\ resources' = [resources\ \mathrm{EXCEPT}\ ![r] = \text{"busy"}]$
$\quad \land\ allocations' = [allocations\ \mathrm{EXCEPT}\ ![t][r] = \mathrm{TRUE}]$

$FreeResource \triangleq \mathrm{CHOOSE}\ r \in ResourceId : resources[r] = \text{"free"}$

$OccupyResource \triangleq$
$\quad \land\ \exists\, r \in ResourceId : resources[r] = \text{"free"}$
$\quad \land\ \mathrm{LET}\ res \triangleq \mathrm{CHOOSE}\ r \in ResourceId : resources[r] = \text{"free"}$
$\quad\quad \mathrm{IN}\ \ resources' = [resources\ \mathrm{EXCEPT}\ ![res] = \text{"busy"}]$
$\quad \land\ \mathrm{UNCHANGED}\ \langle tasks, allocations\rangle$

A task can release a resource when it has finished using it.

$ReleaseResource(t, r) \triangleq$
$\quad \land\ allocations[t][r] = \text{TRUE}$
$\quad \land\ tasks[t] = \text{"running"}$
$\quad \land\ tasks' = [tasks \text{ EXCEPT } ![t] = \text{"waiting"}]$
$\quad \land\ resources' = [resources \text{ EXCEPT } ![r] = \text{"free"}]$
$\quad \land\ allocations' = [allocations \text{ EXCEPT } ![t][r] = \text{FALSE}]$

The next-state relation defines valid state transitions

$ExokernelNext \triangleq$
$\quad \exists\, t \in TaskId,\ r \in ResourceId :$
$\quad\quad (RequestResource(t, r) \lor ReleaseResource(t, r))$

Specification of the overall system behavior

$ExokernelSpec \triangleq ExokernelInit \land \Box[ExokernelNext]_{\langle tasks,\, resources,\, allocations \rangle}$

$vars \triangleq \langle exokernel \rangle$

Fairness $\triangleq$ WF_$vars(TaskAction)$

Invariant: No two tasks can hold the same resource simultaneously

$ResourceExclusivity \triangleq$
$\quad \forall\, r \in ResourceId :$
$\quad\quad \forall\, t1, t2 \in TaskId : (t1 \neq t2) \Rightarrow \lnot(allocations[t1][r] \land allocations[t2][r])$

ASSUME Assumption $\triangleq tasks \in [TaskId \to TaskState]$

THEOREM $Init \Rightarrow$ Fairness
BY DEF $Init$, Fairness
